

Unit-03

Cryptography

3.1 Basic Cryptography

Cryptography is the art and science of securing information to ensure its confidentiality, integrity, authenticity, and non-repudiation. It is a critical component in securing communication, preventing unauthorized access, and safeguarding data in digital environments.

3.1.1 Symmetric Cryptosystems

Symmetric cryptosystems rely on a single key for both encryption (to protect the data) and decryption (to access the data). Both the sender and the receiver must have access to the same secret key, making secure key exchange a fundamental requirement.

- **Examples:** AES (Advanced Encryption Standard), DES (Data Encryption Standard), 3DES.
- **Advantages:**
 - Fast and efficient, making it suitable for encrypting large volumes of data.
 - Simpler algorithms compared to asymmetric cryptography.
- **Disadvantages:**
 - Securely sharing and managing keys is challenging, especially across large networks.
 - Lack of scalability; a unique key is required for every pair of communicating parties.

3.1.2 Public Key Cryptography

Public Key Cryptography, also known as asymmetric encryption, uses a pair of keys:

1. A **public key** to encrypt data, which can be shared openly.
2. A **private key** to decrypt data, which must remain confidential.

This approach eliminates the need for secure key exchange, solving the scalability issue present in symmetric cryptosystems.

- **Examples:** RSA (Rivest-Shamir-Adleman), ECC (Elliptic Curve Cryptography).
- **Advantages:**
 - Ensures secure communication even over unsecured channels.
 - Scalable and ideal for large networks, such as the internet.
- **Disadvantages:**
 - Slower than symmetric encryption due to more complex computations.
 - Computationally expensive for encrypting large datasets.

3.1.3 Cryptographic Checksums

Cryptographic checksums are fixed-length values generated from data using cryptographic hash functions. They ensure the integrity of data by verifying that it has not been tampered with during transmission or storage.

- **How it Works:**
 - A checksum is created by running the data through a hash function like SHA-256, SHA-3, or MD5.
 - If the checksum of the received data differs from the original, it indicates corruption or tampering.
- **Advantages:**
 - Detects accidental errors and intentional alterations.
 - Simple and efficient to compute and verify.
- **Limitations:**
 - Alone, it does not confirm the authenticity of the sender unless combined with a digital signature.
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3.1.4 Digital Signature

A digital signature is a cryptographic technique used to verify the authenticity and integrity of digital messages or documents. It provides strong non-repudiation, ensuring that the sender cannot deny having sent the message.

- **How it Works:**

- The sender generates a hash of the message and encrypts it with their private key to create the digital signature.
- The recipient decrypts the signature using the sender's public key and compares it with the hash of the received message to verify authenticity.

- **Advantages:**

- Confirms the sender's identity, ensuring the message is from a trusted source.
- Detects any alterations to the message after it was signed.

3.1.5 Hashing

Hashing is a process of transforming input data of arbitrary size into a fixed-size output, called a hash or message digest. Cryptographic hash functions ensure that even a minor change in the input produces a vastly different hash, making it reliable for data verification.

- **Examples:** SHA-256, SHA-3, MD5.

- **Advantages:**

- One-way function: The original data cannot be retrieved from the hash.
- Highly sensitive to changes, ensuring tamper detection.

- **Applications:**

- Securing password storage by hashing user passwords in databases.
 - Verifying the integrity of files during transmission or download.
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3.2 Key Management

Key management refers to the process of securely handling cryptographic keys throughout their lifecycle, including generation, exchange, storage, usage, and revocation. It ensures that encryption keys remain confidential and available to authorized entities.

3.2.1 Session and Interchange Keys

- **Session Keys:** Temporary keys used for a single communication session to encrypt data, enhancing security by limiting the scope of key usage.
- **Interchange Keys:** Long-term keys used to securely transfer session keys between communicating parties, typically in public key cryptography systems.

3.2.2 Key Exchange and Generation

- **Key Exchange:** The process of securely sharing encryption keys between two or more parties. Examples include Diffie-Hellman and RSA.
- **Key Generation:** The creation of cryptographic keys using random or pseudo-random algorithms to ensure strong and unpredictable keys.

3.2.3 Cryptographic Key Infrastructures

A framework for managing cryptographic keys, including their creation, distribution, validation, and storage. It often involves a Public Key Infrastructure (PKI) that uses digital certificates issued by trusted Certificate Authorities (CAs).

3.2.4 Storing and Revoking Keys

- **Storing Keys:** Ensuring encryption keys are stored securely using hardware (e.g., HSMs) or encrypted software solutions.
 - **Revoking Keys:** The process of invalidating compromised or outdated keys using Certificate Revocation Lists (CRLs) or Online Certificate Status Protocol (OCSP).
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3.3 Cipher Techniques

Ciphers are algorithms used to convert plaintext into ciphertext for secure communication and vice versa.

3.3.1 Stream and Block Ciphers

- **Stream Ciphers:** Encrypt data one bit or byte at a time in a continuous stream. Suitable for real-time data processing.
- **Block Ciphers:** Encrypt fixed-sized data blocks, ensuring stronger security for bulk data encryption.

3.3.2 Authenticated Encryption (AE)

A cryptographic technique that ensures both confidentiality and integrity of encrypted data. It combines encryption with an authentication tag to verify that the data has not been tampered with during transmission. Examples include AES-GCM and ChaCha20-Poly1305.

3.4 Authentication

Authentication is the process of verifying the identity of a user, system, or entity before granting access to resources. It is a cornerstone of information security, ensuring that sensitive systems and data are accessed only by authorized parties. Modern authentication methods aim to balance ease of use, scalability, and security.

3.4.1 Authentication Basics

Authentication verifies identity using one or more factors:

- **Single-Factor Authentication (SFA):** Relies on one method, such as a password, for validation.
- **Two-Factor Authentication (2FA):** Combines two methods, like a password and a mobile verification code.
- **Multifactor Authentication (MFA):** Involves multiple authentication layers for increased security.

Importance: Strong authentication safeguards against unauthorized access and forms the basis for secure digital interactions.

3.4.2 Passwords

Passwords are the simplest and most widely used authentication method.

- **Best Guidelines for Password Usage:**
 - Change default passwords immediately.
 - Avoid reusing passwords across different systems.
 - Use unique passwords for critical accounts.
- **Limitations:** They can be guessed, stolen, or cracked, requiring additional safeguards.

3.4.3 Password Selection

Creating robust passwords reduces the risk of compromise.

- **Strong Password Features:**
 - Random combinations of letters, numbers, and symbols.
 - Avoidance of sequences or repeated characters.
- **Encouraging Users:** Organizations can enforce password policies, such as minimum length and complexity requirements, to ensure stronger credentials.

3.4.4 Attacking Passwords

Attackers target passwords through various techniques:

- **Social Engineering:** Exploiting human trust to extract passwords.
- **Rainbow Table Attacks:** Using precomputed hash values to reverse-engineer weak passwords.
- **Credential Stuffing:** Testing stolen username-password pairs on multiple sites.

Défense Mechanisms:

- Use salted hashes for password storage.
- Limit login attempts.
- Train users to recognize phishing schemes.

3.4.5 Password Aging

Periodic password updates reduce the likelihood of long-term exploitation.

- **Implementation:** Organizations can set policies to expire passwords after a specific duration.
- **Challenges:** Overly frequent changes can lead to weaker passwords, as users may rely on predictable patterns.

Enhanced Security Measures: Pair password aging with robust password creation guidelines and MFA.

3.4.6 Biometrics

Biometric authentication uses unique biological or behavioural characteristics.

- **Common Biometrics:**
 - **Physiological:** Fingerprints, iris scans, facial recognition.
 - **Behavioural:** Voice recognition, typing patterns.
- **Pros:** Convenient, difficult to replicate.
- **Cons:** Can be affected by environmental conditions, such as lighting or injuries, and raises privacy concerns.

Biometric data is often combined with other methods, such as passwords, for a stronger authentication process.

3.4.7 Multifactor Authentication (MFA)

MFA combines two or more verification methods to enhance security.

- **Types of Factors Used:**
 - Knowledge factors (passwords or PINs).
 - Possession factors (security tokens or OTPs).
 - Inherence factors (biometrics).
- **Benefits:**
 - Provides layered security.
 - Protects accounts even if one factor is compromised.

- **Examples:**

- Online banking: Password + SMS OTP.
- Corporate access: Biometric scan + access card.

Future of MFA: Adoption of passwordless solutions like biometric-based MFA and device recognition for seamless, secure access.

With robust authentication techniques, systems can mitigate risks associated with unauthorized access, data breaches, and identity theft while maintaining usability.
